

19 Setting assumptions

5 primary questions | Oldest to newest

2026 objectives: 4.2.1

11 related questions are filed elsewhere in this volume. Find them by selecting this chapter and enabling related topics in [index.html](#).

Click a question below to begin.

CA1-P1 April 2011 Q6	14 marks
CA1-P1 September 2018 Q4	16 marks
CP1-P1 April 2019 Q3	8 marks
CP1-P1 April 2022 Q7	13 marks
CP1-P1 September 2024 Q6	10 marks

CA1-P1 | April 2011 | Q6

[Current topic](#) | [14 marks total](#) | [Original PDF](#) p4

Related: Ch 7.

- 6** A company which writes general insurance business is developing a new line of insurance policies aimed at university students.
- (i) List the types of insurance it could offer. [5]
- (ii) Outline how the assumptions used to price these policies may differ from those used for the company's existing business. [9]
- [Total 14]

CA1-P1 | September 2018 | Q4

Current topic | 16 marks total | Original PDF p2

September 2018 main paper, recovered from the supplied archive; the extracted folder had contained the spare paper under this date. Related: Ch 18, Ch 20.

- 4** A life insurance company has, to date, written only whole life assurances and immediate annuities. It is about to start writing income protection insurance, to cover only long term sickness or incapacity due to accident or illness.

It intends to price the new income protection business using a multiple state model, and is in the process of setting initial best estimates (i.e. ignoring margins for prudence) of the required assumptions.

- (i) List the main assumptions required for the multiple state model. [4]
- (ii) Discuss data that could be used to set an initial best estimate for each assumption in part (i). [12]

[Total 16]

CP1-P1 | April 2019 | Q3

Current topic | 8 marks total | Original PDF p2

Related: Ch 2, Ch 20.

- 3** Over the last five years in a particular country, the government has been reducing the amount of state healthcare it provides, while the population size has remained unchanged.

An insurance company operating in that country sells a variety of domestic products and is considering the implications of the healthcare reductions for its pricing assumptions.

Discuss the implications for the pricing assumptions of each of the company's products sold in that country:

- (a) Term assurance
- (b) Medically underwritten annuity (where the policyholder provided information on their health status at application stage)
- (c) Private medical care
- (d) Funeral cover (where the insurance company pays the funeral costs for the policyholder).

[8]

CP1-P1 | April 2022 | Q7

[Current topic](#) | [13 marks total](#) | [Original PDF p3](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 7** An actuary has been asked to price a new financial product for an insurance company, using a deterministic model. One of the assumptions in the pricing model is extremely uncertain.

Discuss how the actuary could approach setting this assumption. [13]

CP1-P1 | September 2024 | Q6

Current topic | 10 marks total | Original PDF p3

Related: Ch 23, Ch 26.

- 6** Five dentists are setting up a dental practice and have agreed on the fees to charge for each type of dental treatment. Patients will be expected to pay all relevant fees after each treatment they have received.

One of the dentists has suggested offering patients a dental plan. The dental plan would cover the cost of all treatments for a patient in return for a flat monthly fee. This flat monthly fee would be fixed for 1 year and subject to annual increases thereafter.

- (i) Explain why this dental plan may have been proposed. [2]
- (ii) Describe how the assumptions for pricing the dental plan could be set. [5]
- (iii) Describe the risks to the dental practice of offering this plan. [3]

[Total 10]